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3 April 1992
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8 March 1983

MILITARY SPECIFICATION

MOTOR, ROCKET, ASSEMBLY, M54 OR M123
PARTS FOR

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for the fabrication, assembly and packing of one type of rocket motor parts assembly designated as Motor, Rocket, Assembly, M54 or M123.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-A-2550	-	Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Ammunition, Lot Numbering
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

10048509	-	Rocket Motor, M54
11508955	-	Rocket Motor Assembly, M123
10048520	-	Fin

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11508951	-	Fin
10048519	-	Body, Motor, Rkt, M54
11508954	-	Body, Motor, M123
10048518	-	Spring, Fin

(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117	-	Salt Spray (Fog) Testing, Standard Method of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. Materials used in the fabrication of the motor shall be in accordance with applicable drawings and specifications.

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3.3 Assemblies. The motor assemblies shall comply with all requirements specified on drawing 10048509 or 11508955, and with all requirements specified on applicable specifications.

3.4 Embrittlement. Fin springs manufactured for use in the assembly shall be free from embrittlement as evidenced by not breaking when extended or distorted to the point of minimum permanent distortion (see 4.6.1).

3.5 Heat treatment of motor bodies.

3.5.1 Maximum heat-treatment temperatures. The material used in the fabrication of the motor bodies and the finished motor bodies shall at no time be subjected to temperatures greater than 471° Celsius (C) for 7050 alloy (see 4.6.2). If a salt bath furnace is used for solution heat-treatment, the cold-worked motor bodies shall be pre-heated to relieve internal stresses to preclude the development of cracks during the normal rapid rise in temperature upon introduction into the salt bath.

3.5.2 Grain structure. The motor bodies, after all heat-treatments and aging treatments have been completed, shall not contain eutectic melting, harmful grain boundary conditions or Banding (segregation of inter-metallic compounds) when examined under 200 to 500 power magnification (see 4.6.2).

3.6 Physical strength of motor body.

3.6.1 Yield strength. The portion of the motor body from the threaded end to the nozzle throat, shall not rupture; the threads shall not fail to hold the motor body in the test fixture, and the motor body shall conform to the drawing dimensions after having been subjected to an internal hydrostatic pressure of 14,000 pounds force per square inch (lbf/in^2), 200 or -500 lbf/in^2. The pressure shall be applied while the motor body is not restrained or confined so as to restrict deformation, except at the threads, where deformation shall be limited by the threads of the test fixture and the threads in the test fixture shall represent the least metal conditions of the closure (see 4.6.3).

3.6.2 Ultimate strength. The motor body, from the threaded end to the nozzle throat, shall not rupture nor shall the threads fail while an internal hydrostatic pressure of not less than 14,800 lbf/in^2 is applied in such a manner as not to restrict deformation (see 4.6.4).

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3.7 Protective coating. The exterior and interior surfaces of the motor body, after application of the protective coating, shall not contain greater than 7 total isolated bare spots or pits larger than 0.03125 inch in diameter after the finished motor body has been exposed to a salt spray solution containing 5 percent salt for not less than 168 hours. Fins shall have not greater than 3 total bare spots or pits not greater than 0.03125 inch in diameter after exposure to an identical environment for the same period of time as the motor body (see 4.6.7 and 4.6.8).

3.8 Assemblies.

3.8.1 Fin action. Fins shall move freely, not bend or twist, and shall spring open without hesitation. All fins shall open to within the limits specified on the applicable drawing. Springs, lugs, and eyelets shall show no evidence of distortion, or disassembly, and the number of cracked or split eyelets shall be not greater than the limits specified on the applicable drawing (see 4.6.9).

3.8.2 Physical strength of exit cone. The motor body shall not crack or deform beyond drawing tolerances when stressed within the following limits at the locations specified. Stress readings shall indicate the hoop stress to which the motor exit cone has been subjected. Stress shall be not greater than 80,000 lbf/in^2. Stress at all unspecified exit cone locations shall be not less than those resulting from the application of the specified test equipment, to achieve stress levels not less than those at the specified locations. Location of stress measurement points shall be measured along the longitudinal axis of the motor body beginning at the lip of the exit cone (see 4.6.10). Stress measurement points and minimum stress levels shall be as specified in table I.

TABLE I. Stress measurements.

Measurement points	Minimum levels
0.1875 inch	29,000 lbf/in^2
1.0 inch	16,000 lbf/in^2
1.375 inch	34,000 lbf/in^2

At and beyond the throat and through the remainder of the motor body, stress shall result from not less than 10,000 lbf/in^2 to not greater than 13,000 lbf/in^2 internal pressure.

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3.8.3 Defects, physical. The nozzle exit cone area of the motor body shall be free from all external and internal cracks and metallic discontinuities when examined by "eddy current" equipment adjusted to a sensitivity to disclose metallic discontinuities measuring not greater than 0.062 inch diameter hole (see 4.6.11).

3.8.4 Defects, visual. The motor bodies, after all assembly and machining operations and all other tests, shall be free from deformation beyond drawing tolerances, cracks, surface seams, laps, scratches greater than 0.002 inch deep, or any other metallic discontinuities when visually examined under 3 power magnification (see 4.6.12).

3.8.5 Static firing. The motor bodies shall not fracture or fragment when statically tested with production propellant charges and igniters at not less than 66°C (see 4.6.13).

3.9 Workmanship. All parts and assemblies shall be free of chips, dirt, grease, rust, scale, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All markings shall be clean and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test methods documents and paragraph 4.6 in this specification.

4.4 First article inspection. When specified in the contract or order (see 6.2), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained (see 3.1 and 4.2).

4.4.1 First article sample. Prior to initiating regular production, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.5.5.5 (see 6.3). The first article sample shall consist of:

- 200 complete assemblies for static firing at 65.6°C prior to physical strength test of exit cone.

- 25 motor bodies that have not been subjected to hydrotesting or anodizing.

- 10 sets of components. Motor bodies in this set shall be after hydrotest and anodizing.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change

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occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table II (see 4.2).

4.5.1 Rejection. The first article sample shall be rejected if any component or assembly fails to comply with any of the applicable requirements referenced in 4.5.5. The first article sample shall also be rejected if one or more specimens are classified defective during the ultimate-strength test of motor bodies (see 4.6.4) or if the average burst or failure level of the combined sample is less than 16,500 lbf/in^2. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Embrittlement	3.4	4.6.1
Heat treatment of motor bodies	3.5	4.6.2
Physical strength of motor body	3.6	4.6.3, 4.6.4
Protective coating	3.7	4.6.7, 4.6.8
Assemblies	3.8	4.6.9, 4.6.10
Workmanship	3.9	4.3

4.5.2 Lot formation. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced

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by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, drawing revisions, specifications and specification revisions. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.5.2.1 Data cards. Data cards as specified in the contract or order (see 6.2), shall be prepared for each lot in accordance with MIL-STD-1167.

4.5.3 Sampling. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Inspection sampling shall conform to tables III or IV, or as specified herein.

TABLE III. Sampling plan A.

Lot Size	Sample Size
2 to 32	All
33 to 280	32
281 to 500	48
501 to 3200	73
3201 to 10,000	86
10,001 to 35,000	108
35,001 to 150,000	123
150,001 to 500,000	156
500,001 and over	189
In all cases: Acceptance number is ZERO Rejection number is ONE	

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TABLE IV. Sampling plan B.

Lot Size	Sample Size
2 to 20	All
21 to 280	20
281 to 1200	47
1201 to 3200	53
3201 to 10,000	68
10,001 to 35,000	77
35,001 to 150,000	96
150,001 to 500,000	119
500,001 and over	143
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.4 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall, if specified in the contract or order (see 6.2), supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority shall not be required with subsequent certificates unless, during the course of

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the contract, this authority is vested in another agent of the certifying organization.

4.5.5 Examination. The examination listed in the following subparagraphs shall be performed on inspection lots (see 4.5.1) of the components or assemblies designated in the applicable paragraph headings. Certain major defects, when specified, shall be examined on all.

4.5.5.1 Fin. Drawing 10048520 or 11508951 cover a detail of drawing 10048509 or 11508955 and table V. Refer to drawings listed herein (see 2.1.2).

TABLE V. Classification of defects, fin.

Categories	Defects	Method of Inspection	Sampling Plan
Critical:	None defined.		
Major:			
101	Thickness, minimum	Gage	Table III
102	Diameter of hole, maximum	Gage	Table III

4.5.5.2 Rocket motor body. After anodizing M54 or M123 rocket motors, see drawing 10048519 (M54) or 11508954 (M123) which cover details of drawing 10048509 (M54) or 11508955 (M123) and table VI.

TABLE VI. Classification of defects, motor rocket, body.

Categories	Defects	Method of Inspection	Sampling Plan
Critical:			
1.	Hydrostatic test stamp missing (see 3.6.1)	Visual	100%
Major:			
101.	Concentricity of thread with inside diameter	Gage	100%
102.	Concentricity of throat with inside diameter	Gage	100%

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TABLE VI. Classification of defects, motor rocket, body - (continued).

Categories	Defects	Method of Inspection	Sampling Plan
103.	Concentricity of inside of exit cone with inside diameter	Gage	100%
104.	Longitudinal alignment of slots with centerline of inside diameter	Gage	100%
106.	Pitch diameter, minimum	Gage	100%
107.	Length of thread, minimum	Gage	100%
108.	Lengths of datum diameters in exit cone section	Gage	100%
109.	Wall thickness of cylindrical section, minimum	Gage	100%
110.	Wall thickness at end of exit cone, minimum	Gage	100%
111.	Wall thickness at throat, minimum	Gage	100%
112.	Inside diameter of cylindrical section, minimum	Gage	100%
113.	Inside diameter at throat	Gage	100%
114.	Diameter of eyelet holes, maximum	Gage	100%
115.	Distance from centerline of holes to bottom of slots, minimum	Gage	100%
116.	Width of slots, maximum	Gage	100%
117.	Total length	Gage	100%
118.	Protective coating missing, incomplete or damaged (except for hanger marks)	Visual	100%
119.	Major diameter, minimum	Gage	100%

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4.5.5.3 Spring, fin. See drawing 10048518 which covers details shown on drawing 10048509 (M54) or 11508955 (M123) and table VII.

TABLE VII. Classification of defects, spring, fin.

Categories	Defects	Method of Inspection	Sampling Plan
Critical:	None defined.		
Major:			
101.	Torque inadequate at either angle of deflection	Gage	Table III

4.5.5.4 Assembly. See drawing 10048509 (M54) or 11508955 (M123) and table VIII for further details.

TABLE VIII. Classification of defects, assembly.

Categories	Defects	Method of Inspection	Sampling Plan
Critical:			
1.	Exit cone physical strength test stamp missing (see 3.8.2)	Visual	100%
2.	Exit cone physical defects test stamp missing (see 3.8.3)	Visual	100%
Major:			
101.	Fin action (see 3.8.1 and 4.6.9)	Test	Table IV
102.	Weight, maximum	Gage	Table IV
103.	Diameter across folded fins, maximum	Gage	Table IV
104.	Components insecure, damaged (see 3.8.4), missing or incorrectly assembled	Visual	Table IV

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4.5.5.5 First article testing.

4.5.5.5.1 First article sample. The first article sample shall be examined 100 percent for all of the characteristics in 4.5.5 and then subjected to the tests listed below. Rejection of the first article sample shall be in accordance with 4.2.3.

4.5.5.5.1.1 Embrittlement. Ten of the fin springs shall be tested as specified in 4.6.1 (see 3.4).

4.5.5.5.1.2 Grain structure in motor body. Five of the motor bodies from the first article sample that were not hydrotested or anodized shall be tested as specified in 4.6.2 (see 3.5.1 and 3.5.2).

4.5.5.5.1.3 Yield strength of motor body. The remaining 20 motor bodies from the first article sample that were not hydrotested or anodized shall be tested as specified in 4.6.3 (see 3.6.1).

4.5.5.5.1.4 Ultimate strength of motor body. The 20 motor bodies from 4.5.5.5.1.3 shall be tested as specified in 4.6.4. Samples subjected to this test shall be scrapped (see 3.6.2).

4.5.5.5.1.5 Throat contour of motor body. The 10 motor bodies from the sets of components (hydrotested and anodized) shall be tested as specified in 4.6.5 (see drawing 10048519 (M54) or 11508954 (M123)).

4.5.5.5.1.6 Surface finish of motor body. The 10 motor bodies from 4.5.5.5.1.5 shall be examined for surface finish as specified in 4.6.6 (see drawing 10048519 (M54) or 11508954 (M123)).

4.5.5.5.1.7 Protective coating of motor body and fins.

4.5.5.5.1.7.1 Salt spray method. Six motor bodies from 4.5.5.5.1.6 and 6 fins from the first article sample shall be tested as specified in 4.6.7 (see 3.7).

4.5.5.5.1.7.2 Direct measurement method. Four of the motor bodies from 4.5.5.5.1.6 and 4 of the fins from the first article sample shall be tested as specified in 4.6.8 (see drawing 10048519 (M54) or 11508954 (M123) and drawing 10048520 (M54) or 11508951 (M123).

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4.5.5.5.1.8 Physical strength of exit cone. The 200 complete assemblies shall be tested as specified in 4.6.10 (see 3.8.2).

4.5.5.5.1.9 Physical defects in motor body. The 200 complete assemblies from 4.3.3.1.8 shall be tested as specified in 4.6.11 (see 3.8.3).

4.5.5.5.1.10 Visual defects in motor body. The 200 completed assemblies from 4.5.5.5.1.9 shall be examined as specified in 4.6.12 (see 3.8.4).

4.5.5.5.1.11 Static firing of assembly. The 200 complete assemblies from 4.5.5.5.1.10 shall be tested as specified in 4.6.13 (see 3.8.5).

4.5.5.5.2 Production sample.

4.5.5.5.2.1 Embrittlement. A sample of 10 fin springs shall be selected from each heat-treatment batch and tested as specified in 4.6.1. The heat-treatment batch shall be rejected if one or more springs are classified defective (see 3.4).

4.5.5.5.2.2 Grain structure, and maximum and excessive heat-treatment temperature . A sample of 2 motor bodies shall be selected from each heat-treatment and age-treatment batch and tested as specified in 4.6.2. The batch shall be rejected if one or more bodies are classified defective, or if the temperature recording equipment indicates that the upper temperature limit has been exceeded during any phase of fabrication (see 3.5.1 and 3.5.2).

4.5.5.5.2.3 Yield strength of motor body, critical defect. This test shall be performed 100 percent. The test shall be performed as specified in 4.6.3. Motor bodies classified defective shall be rejected and removed from the lot (see 3.6.1).

4.5.5.5.2.4 Ultimate strength of motor body. A sample of 20 motor bodies shall be selected from each motor body inspection lot (after satisfactory conformance to 4.5.5.5.2.3 and tested as specified in 4.6.4. The test results shall be recorded and copies furnished as specified by the contract (see 6.2). Samples subjected to this test shall be scrapped (see 3.5.2).

4.5.5.5.2.5 Throat contour of motor body, major defect. A sample of 30 motor bodies shall be selected from each inspection lot and tested as specified in 4.6.5. The inspection lot shall be

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rejected if 2 or more bodies are classified defective. If 1 body is classified defective, a second sample of 30 shall be selected from the inspection lot and tested as specified in 4.6.5. Testing for Government acceptance may be stopped and the inspection lot shall be rejected upon encountering 1 defective motor body in this second sample (see drawing 10048519 (M54) or 11508954 (M123).

4.5.5.5.2.6 Surface finish of motor body, major defect. A sample of 32 motor bodies shall be selected from each inspection lot and tested as specified in 4.6.6. The inspection lot shall be rejected if 2 or more bodies are classified defective. If 1 body is classified defective, a second sample of 32 motor bodies shall be selected from the inspection lot and tested as specified in 4.6.6. Testing for Government acceptance may be stopped and the inspection lot shall be rejected upon encountering 1 defective motor body in this second sample (see drawing 10048519 (M54) or 11508954 (M123).

4.5.5.5.2.7 Protection against atmospheric stress corrosion, critical defect. The contractor shall establish process controls and schedules to facilitate the Government's surveillance of the contractor's conformance to assure against exceeding the allowable elapsed time from when any surface of the motor is final machined until the protective coating is applied. All products not adequately protected shall be rejected and removed from the process and scrapped (see drawing 10048519 (M54) or 11508954 (M123).

4.5.5.5.2.8 Protective coating of fins and motor body. In determining the adequacy of the protective finish, the contractor shall select either of the following methods for acceptance testing of regular production of fins and motor bodies for the Government.

4.5.5.5.2.8.1 Salt spray method, major defect. A sample of 3 motor bodies and 3 fins shall be selected from each inspection lot and tested as specified in 4.6.7. The applicable inspection lot shall be rejected if 1 or more of the motor bodies or 1 or more fins are classified defective (see 3.7).

4.5.5.5.2.8.2 Direct measurement method, major defects. A sample of 3 motor bodies and 1 fin shall be selected from each inspection lot and tested as specified in 4.6.8. The inspection lot shall be rejected if 1 or more motor bodies are classified defective, or if the fin is classified defective (see drawing 10048519 (M54) or 11508954 (M123) and drawing 10048520 (M54) or 11508951 (M123).

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4.5.5.5.3 Physical strength of exit cone, critical defect. This test shall be performed 100 percent on final assemblies after completion of all fabrication procedures and all testing, except the testing listed below. This test shall be performed as specified in 4.6.10. Assemblies classified defective shall be rejected and removed from the lot (see 3.8.2).

4.5.5.5.4 Physical defects in motor body, critical defect. This test shall be performed 100 percent on the product acceptable in 4.5.5.5.3. This test shall be performed as specified in 4.6.11. Assemblies classified defective shall be rejected and removed from the lot (see 3.8.3).

4.5.5.5.5 Visual defects in motor body, critical defect. This examination shall be performed 100 percent on the product acceptable in 4.5.5.5.4. This examination shall be performed as specified in 4.6.12. Assemblies classified defective shall be rejected and removed from the lot (see 3.8.4).

4.5.5.5.6 Static firing of assembly, critical defect. A sample of 200 complete assemblies shall be selected from each lot and tested at a Government Proving Ground (see 6.2) as specified in 4.6.13. The lot shall be rejected and testing for Government acceptance shall be stopped upon encountering 1 assembly that is classified defective (see 3.8.5).

4.6 Test methods and procedures. The test methods and procedures described in each of the following tests represent only 1 method and 1 procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using these or substitute methods and procedures. Whichever option is selected, detailed descriptions must be submitted to the Government for approval prior to use. Government approval will be based on an evaluation and demonstration to determine the effectiveness of all of the tests and test procedures (see 4.3).

4.6.1 Embrittlement. The fin spring shall be torqued in a manner simulating the actual use of the spring. The spring helix shall be placed over a steel rod 0.124 + 0.000, - 0.010 inch in diameter and the torque applied and removed to affect increasing angles of deflection until the spring fails to return to its original configuration, as specified in 3.4. The spring shall be classified defective if it breaks or cracks prior to achieving the condition of minimum permanent deformation. Samples subjected to this test shall be scrapped and not returned to any lot.

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4.6.2 Grain structure and maximum temperature. The contractor shall provide automatic time and temperature recording equipment to monitor the applicable sequences of heat-treatment to assure compliance with the specified requirements for maximum temperature, as specified in 3.5.1. To determine presence of microscopic defects, the motor body shall be sectioned through diametrically opposed fin slots and the two halves separated from the cylindrical portion of the body at a point to include approximately 0.5 inch of the cylindrical section with the throat and exit cone sections. The cylindrical sections shall be cut to yield a specimen to include the throat and approximately 0.5 inch of the sidewall. One section from each pair of sections (from the same motor) shall be polished, etched and examined under the required magnification. The motor body shall be classified defective if it fails to conform to any of the specified requirements (see 3.5.2).

4.6.3 Yield strength of motor body. The motor body, after all machining operations have been completed, and prior to anodizing, shall be assembled to the applicable test equipment and subjected to the internal hydrostatic pressure specified for not less than 2 seconds. The medium used in testing may be water, or any liquid compatible with the motor body and test equipment. The test equipment shall automatically apply hydrostatic pressures within the prescribed limits. The test equipment shall automatically stamp the component if the component has actually been subjected to the required pressures. The test equipment shall automatically assure that the component is not stamped if it is not subjected to the lowest level specified and as specified in 3.6.1. The test equipment shall have automatic controls that shall lock the equipment in an inoperative condition at any time that the upper level of test pressure is exceeded. Components subject to pressures that exceed the upper specified limit shall be rejected and not returned to any lot. Repair and adjustments to the test equipment shall follow any occurrence of excessive pressure. After testing, the motor body shall be dried and visually examined for bulges and deformation. Motor bodies that appear to be deformed beyond drawing dimensions, burst or have threads that fail to hold shall be classified defective.

4.6.4 Ultimate strength of motor body. The motor body, after having exhibited acceptable yield strength characteristics, shall be assembled to the applicable destructive test equipment and subjected to an internal hydrostatic pressure of increasing magnitude until the motor body ruptures or until the threads pull out of the test fixture, as specified in 3.6.2. The pressure

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level, when failure occurs, shall be recorded to the nearest 200 lbf/in^2 and the results distributed in accordance with the instructions in 6.2.

4.6.5 Contour of motor body throat. The contour of the throat area shall be determined by use of the profile template type measure equipment. The contour shall be determined from a point approximately 0.5 inch forward of the radium blending into the cylindrical surface to the rearmost datum diameter in the exit cone. The motor body shall be classified defective if it fails to conform to the limits specified on the applicable drawing.

4.6.6 Surface finish of motor body. The surface finish of anodized motor bodies shall be determined by use of "boretracker" type equipment. The surface finish shall be determined by the beginning of the throat approach radius (the radium adjacent to the cylindrical surface) to the rearmost datum diameter of the exit cone. The anodized motor body shall be classified defective if the surface finish fails to conform to the drawing requirement.

4.6.7 Protective coating (salt spray method). The component shall be tested as specified in ASTM B 117 using the required salt concentrations for the required period of time. After testing, the components shall be rinsed in warm water, without rubbing, allowed to dry, without wiping, and visually examined. The component shall be classified defective if the visible corrosion exceeds the specified limits (see 3.7).

4.6.8 Protective coating thickness (direct measurement). The motor body shall be sectioned to yield two halves. The primary surfaces of interest are diametrically opposed longitudinal sections containing longitudinal exposure of the contour of the throat. These sections shall include the throat diameter and approximately 0.25 inch of the approach surface to the throat. One edge of both specimens (surfaces that were not developed in the same cut on the diameter) shall be polished and etched to reveal the protective coating thickness. This thickness shall be measured to the nearest 0.00005 inch. The fin shall be sectioned perpendicular to its length approximately 0.5 inch from the hole. One of the exposed cross-sections shall be polished and etched to reveal the protective coating thickness. This thickness shall be measured to the nearest 0.00005 inch, as specified in 3.7. The component shall be classified defective if any protective coating thickness fails to conform to thickness not less than the requirements specified on the applicable drawing.

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4.6.9 Fin action. The complete assembly, including the fin retainer, shall be placed on an anvil which will locate the rear edge of the exit cone 2.500 inch ± 0.100 inch above the surface of a table and orient the longitudinal axis of the assembly perpendicular to the table. The test equipment shall protect the protective coating from damage during testing. When the assembly is in position for test, the fin retainer is moved towards the threaded end of the motor body thereby releasing the fins. The assembly shall be classified defective if one or more fins fail to fully open (fails to contact the surface which is perpendicular to the longitudinal axis of the assembly and 0.875 ± 0.015 inches to the rear of the lip of the exit cone, as specified in 3.8.1).

4.6.10 Physical strength of exit cone. The complete assembly shall be assembled to the applicable test equipment and required stresses applied. The test equipment shall be functionally verified at not greater than 4 hour intervals of production, prior to and immediately following acceptance testing at the beginning and end of the work shift. This verification shall consist of stressing test standards capable of determining that stresses and intended hydrostatic pressures conform to the specified requirements, as specified in 3.8.2. Stress application measurements shall be centered, longitudinally at the specified locations. The result of each test equipment verification shall be recorded and made available to the Government. The Government shall require and witness a verification during any unscheduled, reasonable period of production. Unscheduled verifications shall be kept to a minimum to minimize interruptions in production. If the functional verification reveals that the test equipment is applying non-conforming stress to the product, disposition of the product test since the last acceptable functional verification shall be as follows:

a. Understressed. If 1 or more stress points are not being adequately stressed, the affected product shall be resubmitted to the test after the required repairs or adjustments have been made to the test equipment.

b. Overstressed. If 1 or more stress points, including the internal hydrostatic pressure, have been subjected to excessive stress levels, the affected product shall be rejected and removed from the lot and scrapped.

4.6.11 Physical defects. The assembly shall be assembled to the applicable test equipment and the entire inner surface of the exit cone shall be scanned by the probe of an eddy current type

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tester using the "phase shift" method of detection. Standard motor bodies (standards) containing the specified metallic discontinuities shall be used to calibrate and adjust the test equipment to produce the proper accept or reject signal, as specified in 3.8.3. The test equipment shall automatically mark the assembly only if no reject area or areas were encountered and shall reliably preclude marking of the assembly if one or more areas of non-conformance are encountered.

4.6.12 Visual defects examination. The motor body of the motor assembly, after the completion of all non-destructive examinations and tests, shall be examined under 3 power magnification over the complete surface of the entire area between the exit cone lip and approximately 0.25 inch down the cylindrical straight sided portion of the pressure area for compliance with visual defects requirement, as specified in 3.8.4. The entire remainder of the complete external surface of the motor shall be subjected to unaided visual examination for compliance with the visual defects requirements. The assembly shall be classified defective if it fails to comply with the visual defects requirement on any surface other than the unmachined lug band surface. A motor containing a blemish on the unmachined lug band surface of a nature or visual appearance of depth, or questionability of ultimate depth, beyond 0.002 inch or beyond the lug band rise thickness, whichever is less, shall be classified defective.

4.6.13 Static functioning. The motor assembly shall be loaded with a production igniter and propellant charge and conditioned at 66 ± 3°C for a period of not less than 12 hours and statically fired within 2 minutes after removal from the conditioning chamber. A pressure time trace shall be obtained to establish if failure was due to a pressure greater than 10,000 lbf/in^2, as specified in 3.8.5. Following firing, the motor shall be visually examined. The motor will be classified defective if visual examination reveals that the motor has fractured or fragmented providing the greatest pressure applied was not greater than 10,000 lbf/in^2.

4.7 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage

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shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Packaging, packing and marking. The motor parts assemblies shall be packaged, packed and marked in accordance with MIL-STD-1169, Level A or B.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. When first article is required (see 3.1 and 4.4)

d. When data cards are required (see 4.5.2.1)

e. When certification of product is required (see 4.5.4)

f. Where static firing will be conducted (see 4.5.5.5.6)

g. Instructions for distribution of results of ultimate strength tests of motor bodies (see 4.5.5.5.2.4 and 4.6.4), and results of first article static firing tests of complete assemblies. The contracting officer will determine distribution of test results

h. Levels of preservation, packing and marking (see 4.7 and section 5).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Engine
Missile engines
Propellant, missile

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Review Activity:
Army - AR

Project No. 1340-A619

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-60003E (MI)	

3. DOCUMENT TITLE
MOTOR, ROCKET, ASSEMBLY, M54 OR M123 PARTS FOR

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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